



Basic Emergency Care

Wounds and Wound healing

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Definition

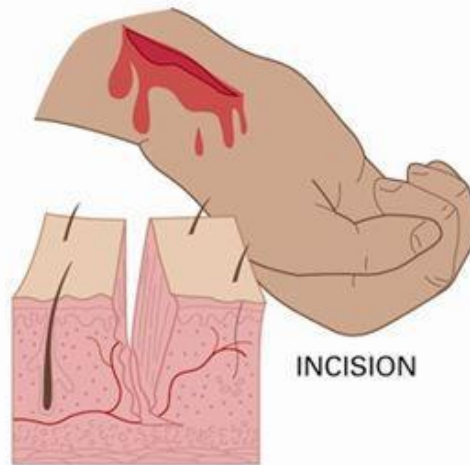
- A wound is a break in the continuity of the skin, the break caused by violence or trauma to the tissue.
- A wound occurs when the integrity of any tissue is compromised (e.g. skin breaks, muscle tears, burns, or bone fractures).
- A wound may be caused by an act, such as a gunshot, fall, or surgical procedure; by an infectious disease; or by an underlying condition.
- ➤ Wounds vary in depth and severity.

Types of wound

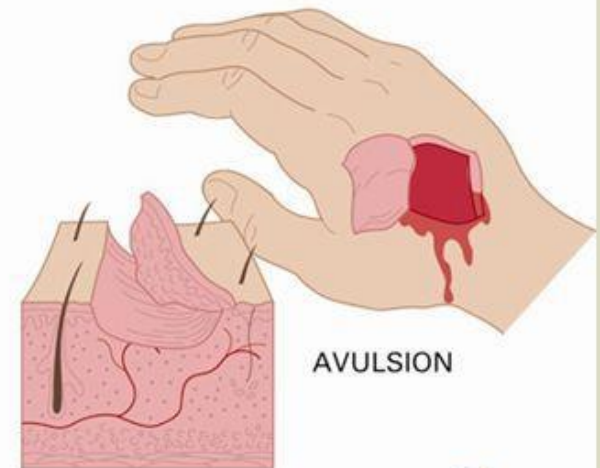


- Abrasion: occurs when your skin rubs or scrapes against a rough or hard surface.
- Laceration: is a deep cut or tearing of skin.
- Puncture: deep narrow tract.
- Avulsion: tissue forcibly torn.
- ➤ Puncture wounds have high infection risk.

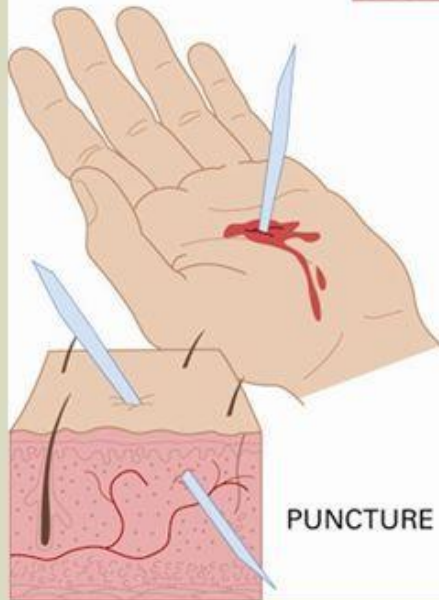
CLASSIFICATION OF OPEN WOUNDS



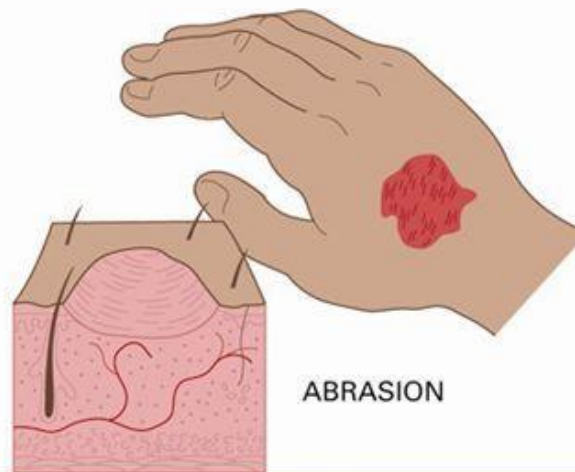
INCISION



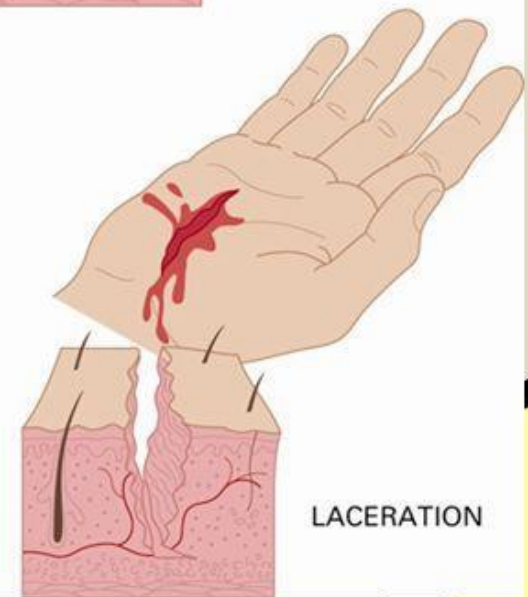
AVULSION



PUNCTURE



ABRASION



LACERATION

Closed vs open wound

- Open wound ----broken skin and exposed body tissue
- Closed wound ----injured tissue not exposed (blunt trauma),damage to underlining tissue
- ➤ Closed wounds may hide deeper injury.

Classification of wound

- **Clean:** They are uninfected, no inflammation is present, and are primarily closed, these wounds do not enter respiratory, alimentary, genital, or urinary tracts.
 - **clean-contaminated:** These wounds lack unusual contamination enter the respiratory, alimentary, genital, or urinary tracts. However, these wounds have entered these tracts under controlled conditions.
 - **Contaminated:** These are fresh, open wounds that can result from insult to sterile techniques or leakage from the gastrointestinal tract into the wound.
 - **dirty-infected:** These wounds typically result from improperly cared for traumatic wounds. demonstrate devitalized tissue.
- ➤ Early cleaning prevents wound infection.

Tidy vs untidy wounds

- **Tidy wounds:** These are inflicted by sharp instruments and contain no devitalized tissues. These wounds can be closed primarily with the expectation of primary healing
- **Untidy wounds** results from crushing, tearing, avulsion, vascular injury or burns and contain devitalized tissues.
- ➤ Untidy wounds may require delayed closure.

Acute vs chronic

CLASSIFICATION OF WOUNDS



ACUTE Recent wound which has yet to progress through the sequential stages of healing

Cuts, Abrasion, Lacerations, Contusions, Puncture, Skin flaps and Bites

BENBOW (2005)



CHRONIC Wound that has arrested in one of the wound healing stages usually inflammatory phase

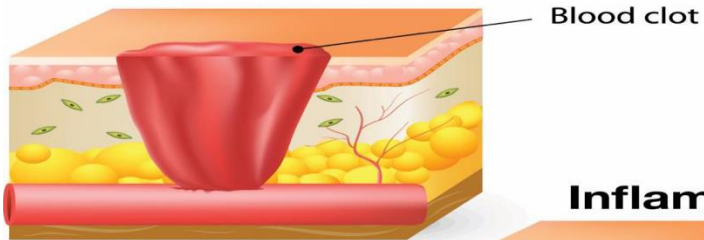
Any wounds > 3 months considered

Stages of wound healing

- **Hemostasis** - After the blood vessels narrow, slowing blood loss, platelets stick together and make a platelet plug at the site of the wound. Then coagulation occurs as fibrin makes a kind of mesh trap and platelets and red blood cells get caught and form a clot.
- **Inflammation** - The body begins cleaning the wound. Debris is taken away from the site to prevent infection.
- **Proliferative** - The body begins to make new blood vessels and new skin is made.
- **Maturation and re-modification** - Other damaged cells are repaired, including nerve cells.
- • ➤ Scar strength improves over months.

WOUND HEALING

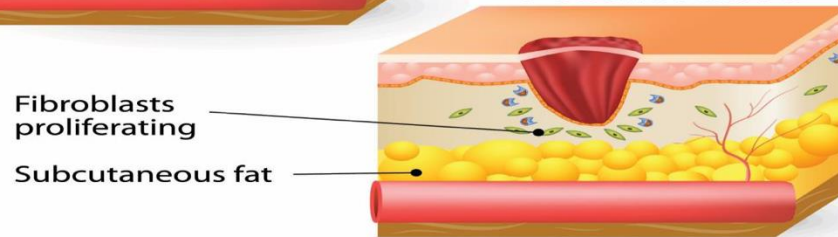
Hemostasis



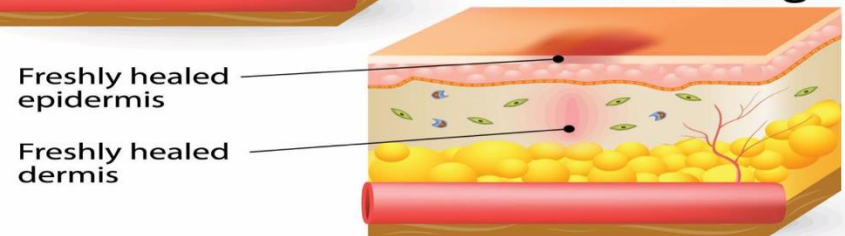
Inflammatory



Proliferative



Remodeling



Factors Affecting Healing

- Diabetes, smoking, malnutrition
- Ischemia, infection
- Foreign bodies
- ► Good nutrition accelerates healing.

Acute wound management

- Stopping of bleeding (elevation, pressure)
- Cleansing
- Exploration
- Debridement
- Repair
- Replacement of lost tissue
- Skin cover if required
- Wound
- ➤ High-pressure irrigation removes contaminants.

Closure Types

- • Primary closure
- • Delayed closure
- • Secondary intention
- • ► Choose closure type based on contamination.

When to Seek Medical Care

- • Deep wounds
- • Exposed bone/tendon
- • Signs of infection
- • ► Bite wounds always require medical review.

Signs of Infection

- • Redness, swelling
- • Warmth, pus, odor
- • Fever
- • ► Early antibiotics prevent complications.

High-Risk Wounds

- • Animal bites
- • Deep punctures
- • Farm injuries
- • ➤ These wounds need immediate evaluation.

Wound Care Do's & Don'ts

- • DO clean gently
- • DO cover with sterile dressing
- • DON'T use cotton inside wounds
- • ► Proper technique prevents infection.